Computer Architectures
Exam of 3.2.2026 - part I – Group C

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Question #1

Speaking about VLIW (Very Long Instruction Word) architectures, you are requested to

- Describe their key characteristics in terms of hardware
- Describe the type of code running on them
- List their advantages and disadvantages.

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Question #2

Let consider a superscalar RISC-V architecture including the following functional units working in single-precision FP (for each unit the number of clock periods to complete one instruction is reported):

- Integer ALU: 1 clock period
- Data memory: 1 clock period
- FP arithmetic unit: 2 clock periods (pipelined)
- FP multiplier unit: 4 clock periods (pipelined)
- FP divider unit: 6 clock periods (unpipelined)

You should also assume that

- The branch delay slot corresponds to 1 clock cycle, and the branch delay slot is not enabled
- Data forwarding is enabled
- The EXE phase can be completed out-of-order.

You should consider the following code fragment and, filling the following tables, determine the pipeline behavior in each clock period, as well as the total number of clock periods required to execute the fragment. The value of the constants k1, k2, k3 and k4 are written in f10, f11, f12 and f13 before the beginning of the code fragment. Registers x1, x2, x3 and x4 are supposed to store the addresses of vectors v1, v2, v3 and v4.

```
; ******************** RISC- **********************
;  for (i = 0; i < 100; i++) {
;
     v4[i] = ((v1[i]+k1)/k3 + (v2[i]+k2)/k4)*v3[i];
;  }
```

		Comments	Clock cycles
	.data		
v1:	.word “100 values”		
v2:	.word “100 values”		
v3:	.word “100 values”		
v4:	.word “100 values”		
	.text		
	addi x5,x0,100	x5← 100	
loop:	flw f1,0(x1)	f1 ← v1[i]	
	fadd f2, f1, f10	f2 ← v1[i]+k1	
	fdiv f3, f2, f12	f3 ← (v1[i]+k1)/k3	
	flw f4,0(x2)	f4 ← v2[i]	
	fadd f5, f4, f11	f5 ← v2[i]+k2	
	fdiv f6,f5,f13	f6 ← (v2[i]+k2)/k4	
	flw f7,0(x3)	f7 ← v3[i]	
	fmul f8,f6,f7	f8 ← (v2[i]+k2)/k4*v3[i]	
	fadd f9,f3,f8	f9 ← (v1[i]+k1)/k3+ (v2[i]+k2)/k4*v3[i]	
	fsw f9,0(x4)	v4[i] ← f9	
	addi x1,x1,4	x1 ← x1 + 4	
	addi x2,x2,4	x2 ← x2 + 4	
	addi x3,x3,4	x3 ← x3 + 4	

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addi x4,x4,4	x4 ← x4 + 4	
addi x5,x5,-1	x5 ← x5 - 1	
bnez x5,loop	branch	
nop		
total		

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addi x5,x0,100																																																										
loop: flw f1,0(x1)																																																										
fadd f2, f1, f10																																																										
fdiv f3, f2, f12																																																										
flw f4,0(x2)																																																										
fadd f5, f4, f11																																																										
fdiv f6,f5,f13																																																										
flw f7,0(x3)																																																										
fmul f8,f6,f7																																																										
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addi x4,x4,4																																																										
addi x5,x5,-1																																																										
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